

Model attacks

Ganghua Wang

May 24, 2022

1 Introduction

Attacks	Change	Goal
Stealing	N/A	Steal functionality
Adversarial	New query	Fool trained model with a new input
Poisoning	Training data	Corrupt the trained model
Inversion	N/A	Compromise user information

Table 1: Comparison of different model attacks.

Model stealing attacks. Figure 1 The attacker’s goal in the model stealing attack, also named model extraction attack, is to infer a trained model. Based on the adversary’s capability, it can be either a white-box scenario that the attacker knows the model structure and may be interested in recovering the model parameters(Tramèr et al., 2016; Wang and Gong, 2018); or a black-box scenario that the attacker can only access the model responses and wants to duplicate or steal the model functionality, i.e., reconstruct a model has comparable accuracy(Orekondy et al., 2019).

Model poisoning attack. Figure 2 The model poisoning attack is the attack that manipulates the training process and causes the defender to learn a corrupted model. This attack is natural in the distributed learning, such as federated learning (Konečný et al., 2016; Chen et al., 2017), where the central server trains a model using the data sent from the clients and the malicious client can manipulate its data to mislead the server. For federated learning, the ‘data’ sent to the server is actually the local model. Therefore, the adversary can either directly change the local model parameters, which is named ‘model poisoning attack’ (Fang et al., 2020), or indirectly influence the local model by modifying the training data, which is named ‘data poisoning attack’ (Jagielski et al., 2018). Be aware that we name all of them as ‘model poisoning attack’ since the ultimate goal is to corrupt the learned model, while the naming of ‘model/data poisoning’ in federated learning corresponds to what we manipulated. Additionally, poisoning attack can be divided into the un-targeted attack that aims to degrade the model accuracy on all inputs and the targeted attack that aims at inputs with specific patterns. The back-door attack (Chen et al., 2017) is one kind of targeted attack. (Compared with the model stealing attack, the defender there becomes the attacker in the model poisoning attack and vice versa, while the framework remains similar.)

Figure: Machine Learning as a Service (MLaaS) and the model stealing attack

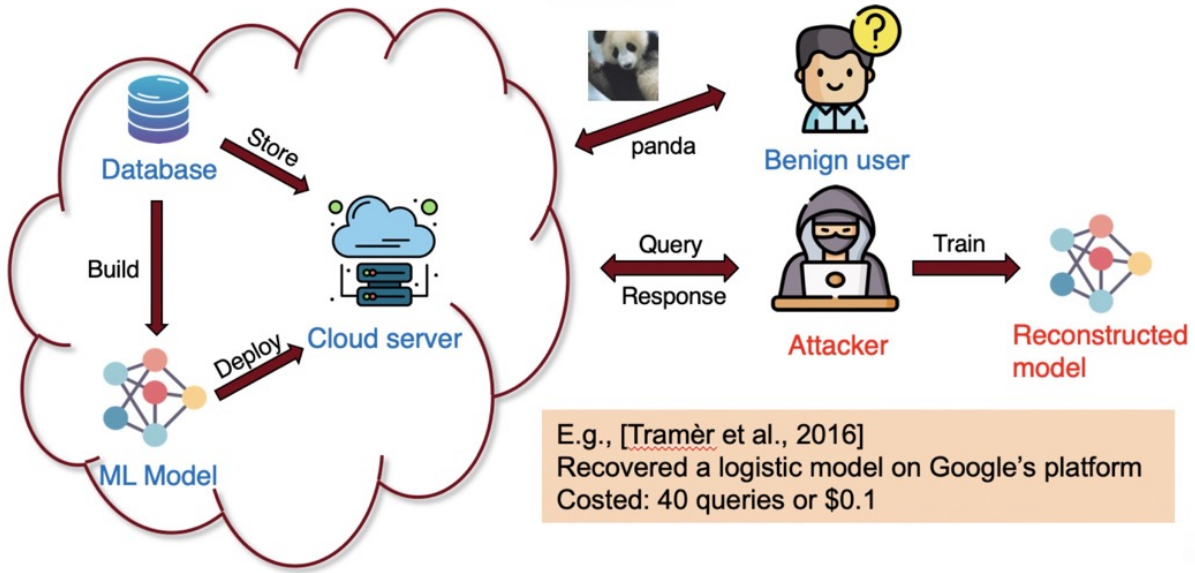


Figure 1: Model stealing/extraction attacks.

Adversarial/evasion attack. Figure 3 For adversarial or evasion attacks, the attacker deliberately adds a carefully designed noise to the inputs/queries to fool the trained model. The perturbed input is called an adversarial example. Usually, the noise is often tiny and human imperceptible, so we cannot distinguish the adversarial example from a benign example. However, the model will respond significantly differently to the adversarial example and the clean input. For example, Lowd and Meek (2005) showed that after changing several words in a spam email, we could avoid the detection of a spam filter. This model instability also raises great interest and discussions in the study of deep neural networks since Szegedy et al. (2013).

Model inversion/reverse-engineering attack. Figure 4 Model inversion attack, also known as model reverse-engineering, focuses on user data privacy instead of the model itself. It aims to infer the sensitive information about the user based on model predictions, such as membership-inference(Shokri et al., 2017) that tries to identify whether the user is in the database and recover the exact values of some attributes(Fredrikson et al., 2014). For example, Fredrikson et al. (2015) showed that the face images used to train a face recognition system could be recovered given people's names only.

References

- Xinyun Chen, Chang Liu, Bo Li, Kimberly Lu, and Dawn Song. Targeted backdoor attacks on deep learning systems using data poisoning. *arXiv preprint arXiv:1712.05526*, 2017.
- Minghong Fang, Xiaoyu Cao, Jinyuan Jia, and Neil Gong. Local model poisoning attacks to

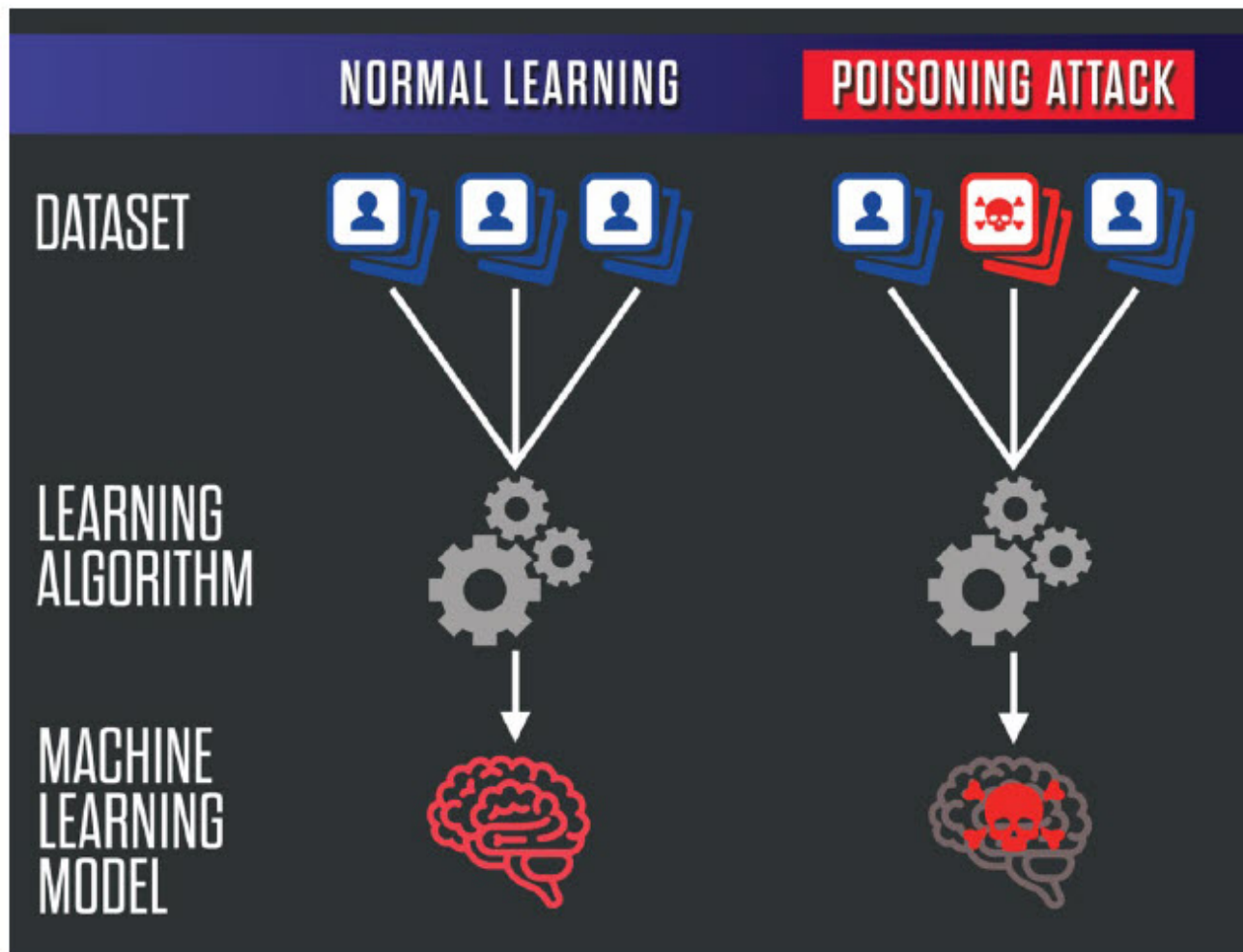


Figure 2: Poisoning attack.

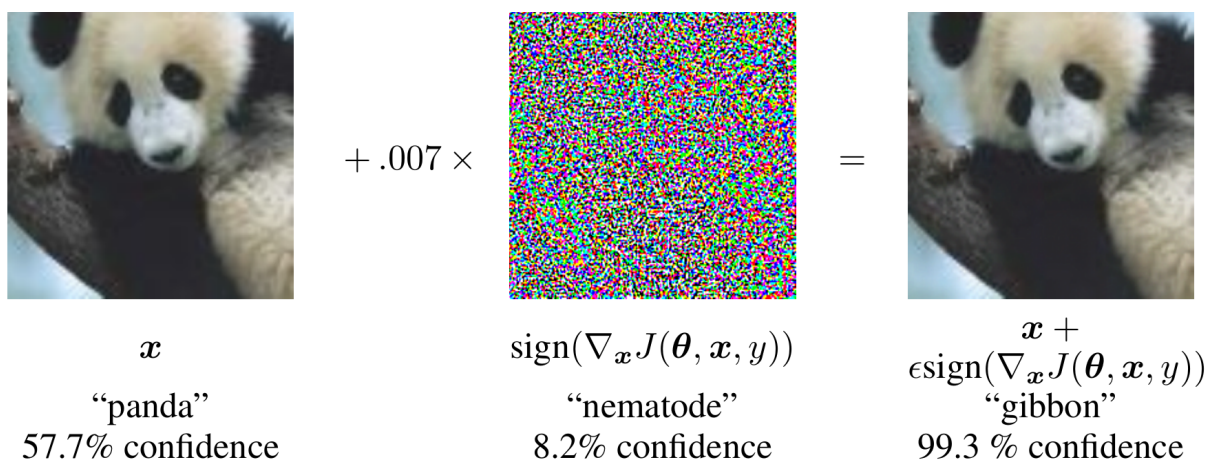


Figure 3: Adversarial/Evasion attack.

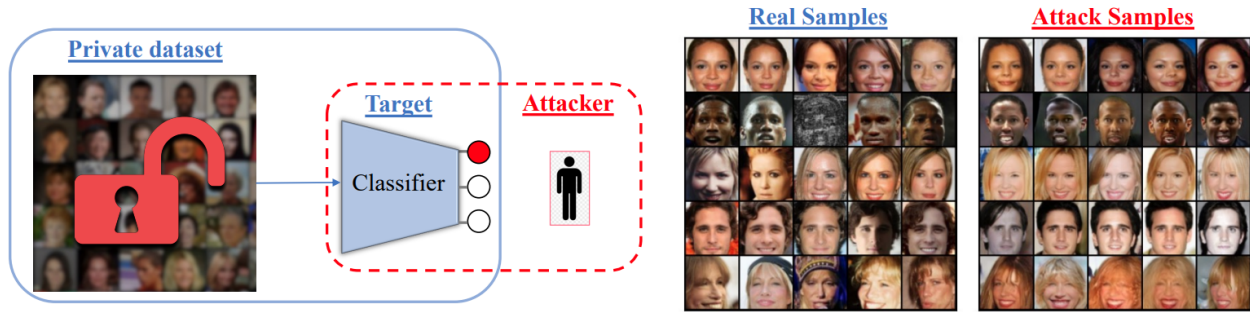


Figure 4: Inversion attack.

{Byzantine-Robust} federated learning. In *29th USENIX Security Symposium (USENIX Security 20)*, pages 1605–1622, 2020.

Matt Fredrikson, Somesh Jha, and Thomas Ristenpart. Model inversion attacks that exploit confidence information and basic countermeasures. In *Proc. ACM SIGSAC*, pages 1322–1333, 2015.

Matthew Fredrikson, Eric Lantz, Somesh Jha, Simon Lin, David Page, and Thomas Ristenpart. Privacy in pharmacogenetics: An {End-to-End} case study of personalized warfarin dosing. In *23rd USENIX Security Symposium (USENIX Security 14)*, pages 17–32, 2014.

Matthew Jagielski, Alina Oprea, Battista Biggio, Chang Liu, Cristina Nita-Rotaru, and Bo Li. Manipulating machine learning: Poisoning attacks and countermeasures for regression learning. In *2018 IEEE Symposium on Security and Privacy (SP)*, pages 19–35. IEEE, 2018.

Jakub Konečný, H Brendan McMahan, Felix X Yu, Peter Richtárik, Ananda Theertha Suresh, and Dave Bacon. Federated learning: Strategies for improving communication efficiency. *arXiv preprint arXiv:1610.05492*, 2016.

Daniel Lowd and Christopher Meek. Adversarial learning. In *Proceedings of the eleventh ACM SIGKDD international conference on Knowledge discovery in data mining*, pages 641–647, 2005.

Tribhuvanesh Orekondy, Bernt Schiele, and Mario Fritz. Knockoff nets: Stealing functionality of black-box models. In *Proc. CVPR*, pages 4954–4963, 2019.

Reza Shokri, Marco Stronati, Congzheng Song, and Vitaly Shmatikov. Membership inference attacks against machine learning models. In *2017 IEEE symposium on security and privacy (SP)*, pages 3–18. IEEE, 2017.

Christian Szegedy, Wojciech Zaremba, Ilya Sutskever, Joan Bruna, Dumitru Erhan, Ian Goodfellow, and Rob Fergus. Intriguing properties of neural networks. *arXiv preprint arXiv:1312.6199*, 2013.

Florian Tramèr, Fan Zhang, Ari Juels, Michael K Reiter, and Thomas Ristenpart. Stealing machine learning models via prediction apis. In *Proc. USENIX Security*, pages 601–618, 2016.

Binghui Wang and Neil Zhenqiang Gong. Stealing hyperparameters in machine learning. In *Proc. IEEE SP*, pages 36–52. IEEE, 2018.